

Cyber Security

Mobile Device Security

By
Dr. Mudassar Raza
Professor
Department of Computer Science
Namal University Mianwali



Mobile Device Security

- Today almost everyone uses a smartphone.
- These devices store our personal data, banking apps, emails, and even office information.
- Because of this, mobile devices have become a major target for cyber attackers.
- In this lecture, we will study the security risks related to mobile devices and how to protect them.

Organization

- **Quick Overview of Mobile Devices**
- Mobile Threats and Attacks
- Countermeasures

Overview of Mobile Devices

- Mobile *computers*:
 - Mainly smartphones, tablets
 - Sensors: GPS, camera, accelerometer, etc.
 - Computation: powerful CPUs (≥ 1 GHz, multi-core)
 - Communication: cellular/4G, 5G, Wi-Fi, near field communication (NFC), etc.
- Many connect to cellular networks: *billing system*
- Statista: Worldwide mobile devices reached 17.7 billion units by 2024. [1]



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Mobile Threats and Attacks

- Mobile devices make attractive targets:
 - People store much personal info on them: email, calendars, contacts, pictures, etc.
 - Sensitive organizational info too...
 - Can fit in pockets, easily lost/stolen
 - Built-in billing system: SMS/MMS (mobile operator), in-app purchases (credit card), etc.
 - Many new devices have near field communications (NFC), used for **contactless payments**, etc.
 - Your **device becomes your credit card**
 - **Location privacy** issues
- NFC-based billing system vulnerabilities

Mobile Device Loss/Theft

- Many mobile devices lost, stolen each year
 - 113 cell phones lost/stolen every minute in the U.S. [15]
 - 56% of us misplace our mobile phone or laptop each month [15]
 - Lookout Security found \$2.5 billion worth of phones in 2011 via its Android app [16]
 - Symantec placed 50 “lost” smartphones throughout U.S. cities [17]
 - 96% were accessed by finders
 - 80% of finders tried to access “sensitive” data on phone

Device Malware

- iOS malware: very little
- Juniper Networks: Major increase in Android malware from 2010 to 2011 [18]
- Android malware growth keeps increasing (\$\$\$)
- Main categories: [19]
 - Trojans
 - Monitoring apps/spyware
 - Adware
 - Botnets

5 Ways to Malware

- **Downloading malicious apps**
 - Most common issue
- **Using a mobile device with an operating system vulnerabilities**
 - Keep OS up to date - Like patching OS on servers.
- **Opening suspicious emails**
 - Phishing, etc. applies here too
 - Employees are people ☹
- **Using non-secure Wi-Fi/URLs**
 - Web browser attacks - Attacks more common on android devices.
 - Current version of whatever browser you use.
- **Receiving text message/voicemail phishing**
 - Hackers often use this information to steal whatever data they can (e.g., ssn, credit card)

Tips to Secure Mobile Devices

- **Don't jailbreak** your device - Removes a lot of its built-in security.
- VPN (virtual private network) - secure “**tunnel**” that access/share info securely over public Wi-Fi networks.
- Download apps only from reputable sources
- Encrypt your data
- Mobile vulnerability scanning
- Update software and hardware
- Train employees
- Have mobile device policies in place

Location Disclosure

- MAC, Bluetooth Addresses, IMEI, IMSI etc. are globally unique
- Infrastructure based mobile communication
- Peer-t-Peer ad hoc mobile communication

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Mobile Access Control

- Very **easy** for attacker **to control a mobile device** if he/she has **physical access**
 - Especially if there's no way to authenticate user
 - Then **device can join botnet**, send SMS spam, etc.
- Need access controls for mobile devices
 - Authentication, authorization, accountability
 - Authentication workflow:
 - Request access
 - Supplication (user provides identity, e.g., John Smith)
 - Authentication (system determines user is John)
 - Authorization (system determines what John can/cannot do)

Authentication: Categories

- Authentication generally based on:
 - Something supplicant knows
 - Password/passphrase
 - Unlock pattern
 - Something supplicant has
 - Magnetic key card
 - Smart card
 - Token device
 - Something supplicant is
 - Fingerprint
 - Retina scan

Authentication: Passwords

- Cheapest, easiest form of authentication
- Works well with most applications
- Also the weakest form of access control
 - Lazy users' passwords: *1234*, *password*, *letmein*, etc.
 - Can be defeated using dictionary, brute force attacks
- Requires administrative controls to be effective
 - Minimum length/complexity
 - Password aging
 - Limit failed attempts

Authentication: Smart Cards/Security Tokens

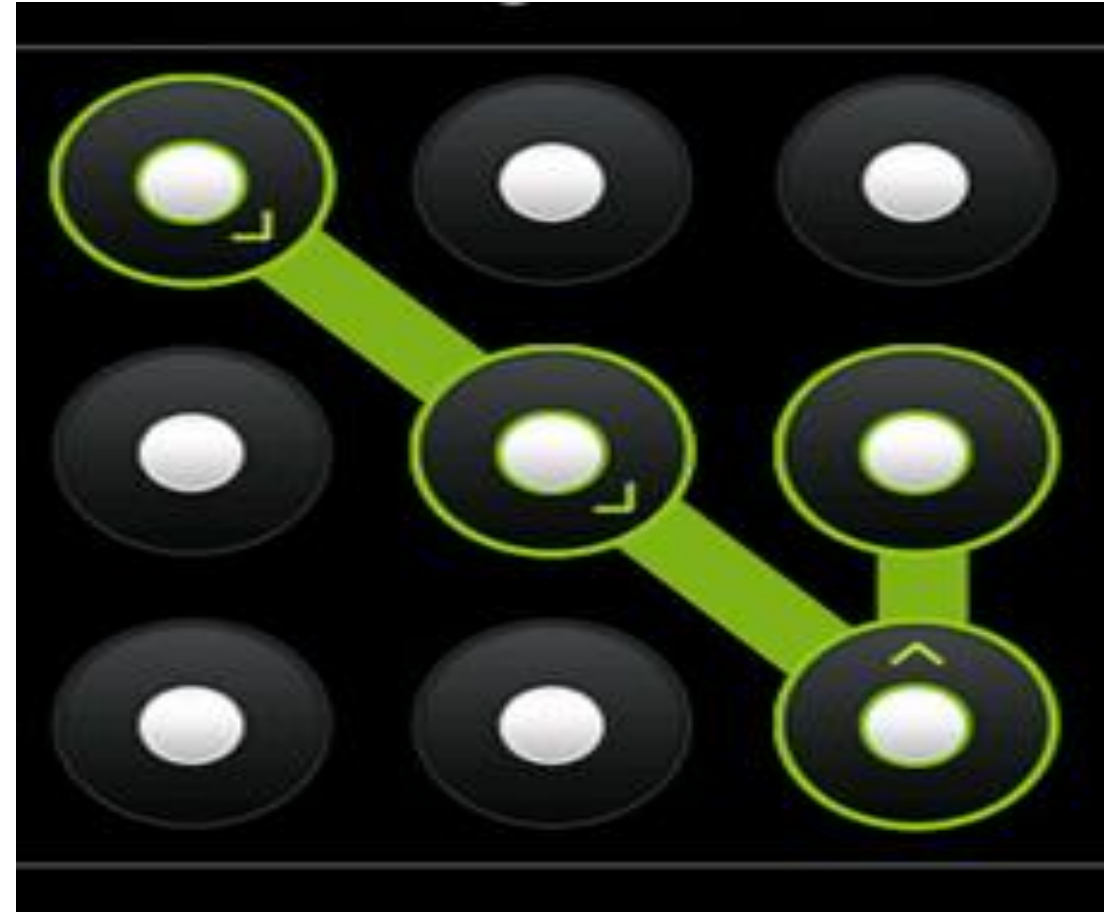
- More expensive, harder to implement
- Vulnerability: prone to loss or theft
- Very strong when combined with another form of authentication, e.g., a password
- Does not work well in all applications
 - Try carrying a smart card in addition to a mobile device!

Authentication: Biometrics

- More expensive/harder to implement
- Prone to error:
 - False negatives: not authenticate authorized user
 - False positives: authenticate unauthorized user
- Strong authentication when it works
- Does not work well in all applications
 - Fingerprint readers becoming more common on mobile devices (Atrix 4G)

Authentication: Pattern Lock

- Swipe path of length 4–9 on 3 x 3 grid
- Easy to use, suitable for mobile devices
- Problems: [30]
 - 389,112 possible patterns; (456,976 possible patterns for 4-char case-insensitive alphabetic password!)
 - Attacker can see pattern from finger oils on screen



Authentication: Comparison

	Passwords	Smart Cards	Biometrics	Pattern Lock
Security	Weak	Strong	Strong	Weak
Ease of Use	Easy	Medium	Hard	Easy
Implementation	Easy	Hard	Hard	Easy
Works for phones	Yes	No	Possible	Yes

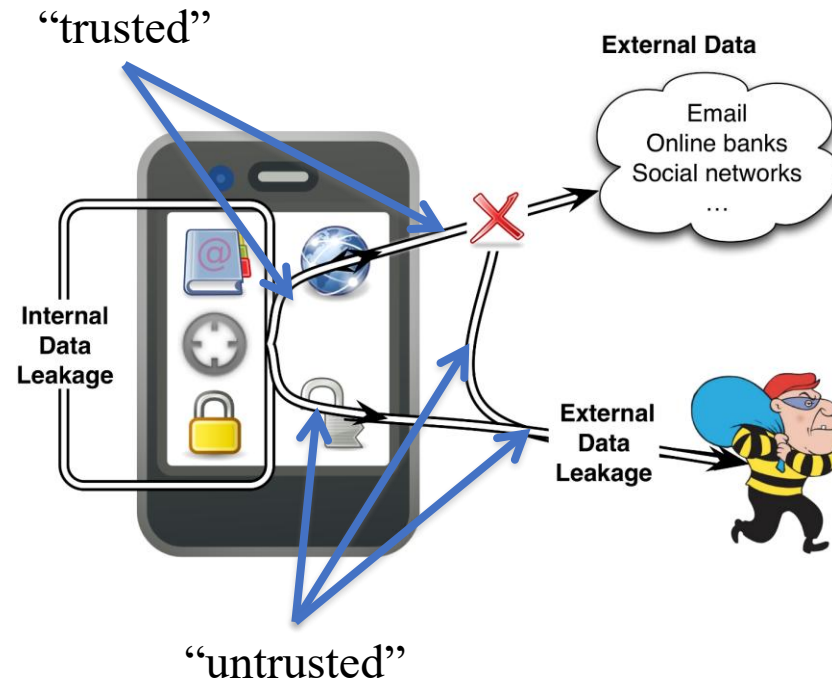
– Deeper problem: mobile devices are designed with single-user assumption...

Mobile Device Information Leakage

- Types of mobile device information sources:
 - Internal to device (e.g., GPS location, IMEI, etc.)
 - External sources (e.g., Chase Bank, etc.)
- Third-party mobile apps can leak info to external sources [32]
 - Send out device ID (IMEI/EID), contacts, location, etc.
 - Apps ask permission to access such info; users can ignore!
 - Apps can intercept info sent to a source, send to different destination!
- Motives:
 - Monitor employees' activity using accelerometers (cited in [32])
 - Ads, market research (include user location, behavior, etc.)
- How do we protect against such information leakage?

Information Flow Tracking (IFT)

- IFT tracks each information flow among internal, external sources
 - Each flow is *tagged*, e.g., “untrusted”
 - Tag propagated as information flows among internal, external sources
 - **Sound alarm if data sent to third party**
- Challenges
 - Reasonable runtime, space overhead
 - Many information sources



Information leakage on mobile devices

Location Privacy Protection

- Strong regulation
 - Corporate
 - Individual
- Dynamic MAC and Bluetooth addresses?
 - Collision
 - How often to change?
- Proxy-based communications
 - Dummy device as proxy
 - Group communications

Device Search and Seizure

- *People v. Diaz*: if you're arrested, police can search your mobile device without warrant [26]
 - Rationale: prevent perpetrators destroying evidence
 - Quite easy to break the law (overcriminalization) [27]
 - Crime severity: murder, treason, etc. vs. unpaid citations
 - "Tens of thousands" of offenses on the books [26]
 - Easy for law enforcement to extract data from mobile devices (forensics) [28]

Summary

- Mobile devices are increasingly popular
- There are many threats and attacks against mobile devices, e.g., loss/theft, sensitive information leakage, and location privacy compromise
- Mobile access control, information leakage protection, and location privacy protection, etc.

References (1)

1. Statista.com (2021). Forecast number of mobile devices worldwide from 2020 to 2024 (in billions)*. Retrieved May 22, 2021 from <https://www.statista.com/statistics/245501/multiple-mobile-device-ownership-worldwide/>
 2. Samsung, “Exynos 5 Dual,” 2012, <http://www.samsung.com/global/business/semiconductor/product/application/detail?productId=7668&iad=2341>
 3. Nielsen Co., “Two Thirds of All New Mobile Buyers Now Opting for Smartphones,” 12 Jul. 2012, http://blog.nielsen.com/nielsenwire/online_mobile/two-thirds-of-new-mobile-buyers-now-opting-for-smartphones/
 4. K. De Vere, “iOS leapfrogs Android with 410 million devices sold and 650,000 apps,” 24 Jul. 2012, <http://www.insidemobileapps.com/2012/07/24/ios-device-sales-leapfrog-android-with-410-million-devices-sold/>
 5. K. Haslem, “Macworld Expo: Optimised OS X sits on ‘versatile’ Flash,” 12 Jan. 2007, Macworld, <http://www.macworld.co.uk/ipod-itunes/news/index.cfm?newsid=16927>
 6. Wikipedia, “iOS,” updated 2012, <http://en.wikipedia.org/wiki/iOS>
 7. Apple Inc., “iPhone Developer University Program,” <http://developer.apple.com/iphone/program/university.html>
 8. Apple Inc, “iOS Security,” http://images.apple.com/ipad/business/docs/iOS_Security_May12.pdf
 9. Android Open Source Project, “Android Security Overview,” <http://source.android.com/tech/security/index.html>
- Presentation organization inspired by T. Eston, “Android vs. iOS Security Showdown,” 2012, <http://www.slideshare.net/agent0x0/the-android-vs-apple-ios-security-showdown>

References (2)

10. A. Rubin, 15 Feb. 2012, <https://plus.google.com/u/0/112599748506977857728/posts/Btey7rJBaLF>
11. H. Lockheimer, "Android and Security," 2 Feb. 2012, <http://googlemobile.blogspot.com/2012/02/android-and-security.html>
12. Android Open Source Project, <http://developer.android.com/about/dashboards/index.html>
13. M. DeGusta, "Android Orphans: Visualizing a Sad History of Support," 26 Oct. 2011, <http://theunderstatement.com/post/11982112928/android-orphans-visualizing-a-sad-history-of-support>
14. <http://opensignalmaps.com/reports/fragmentation.php>
15. [http://www.micro-trax.com/statistics`](http://www.micro-trax.com/statistics)
16. Lookout, Inc., "Mobile Lost and Found," 2012, <https://www.mylookout.com/resources/reports/mobile-lost-and-found/>
17. K. Haley, "Introducing the Smartphone Honey Stick Project," 9 Mar. 2012, <http://www.symantec.com/connect/blogs/introducing-symantec-smartphone-honey-stick-project>
18. Juniper Networks, Inc., "Global Research Shows Mobile Malware Accelerating," 15 Feb. 2012, <http://newsroom.juniper.net/press-releases/global-research-shows-mobile-malware-accelerating-nyse-jnpr-0851976>

References (3)

19. F-Secure, "Mobile Threat Report Q2 2012," 7 Aug. 2012, <http://www.slideshare.net/fsecure/mobile-threat-report-q2-2012>
20. <http://nakedsecurity.sophos.com/2012/04/12/android-malware-angry-birds-space-game/>
21. Via Forensics LLC, "Forensic Security Analysis of Google Wallet," 12 Dec. 2011, <https://viaforensics.com/mobile-security/forensics-security-analysis-google-wallet.html>
22. Proxmark, <http://www.proxmark.org/>
23. libnfc, <http://www.libnfc.org>
24. D. Goodin, "Android, Nokia smartphone security toppled by Near Field Communication hack," 25 Jul. 2012, <http://arstechnica.com/security/2012/07/android-nokia-smartphone-hack/>
25. B. Andersen, "Australian admits creating first iPhone virus," 10 Nov. 2009, <http://www.abc.net.au/news/2009-11-09/australian-admits-creating-first-iphone-virus/1135474>
26. R. Radia, "Why you should always encrypt your smartphone," 16 Jan. 2011, <http://arstechnica.com/gadgets/2011/01/why-you-should-always-encrypt-your-smartphone/>
27. Heritage Foundation, "Solutions for America: Overcriminalization," 17 Aug. 2010, <http://www.heritage.org/research/reports/2010/08/overcriminalization>
28. Wikipedia, http://en.wikipedia.org/wiki/Mobile_device_forensics
29. C. Quentin, <http://www.slideshare.net/cooperq/your-cell-phone-is-covered-in-spiders>

References (4)

- 30. A. J. Aviv, K. Gibson, E. Mossop, M. Blaze, and A. M. Smith, “Smudge Attacks on Smartphone Touch Screens,” Proc. USENIX WOOT, 2010.
- 31. X. Ni, Z. Yang, X. Bai, A. C. Champion, and Dong Xuan, “DiffUser: Differentiated User Access Control on Smartphones,” *Proc. IEEE Int’l. Workshop on Wireless and Sensor Networks Security (WSNS)*, 2009.
- 32. W. Enck, P. Gilbert, B.-G. Chun, L. P. Cox, J. Jung, P. McDaniel, and A. N. Sheth, “TaintDroid: An Information-Flow Tracking System for Realtime Privacy Monitoring on Smartphones,” Proc. USENIX OSDI, 2010, <http://appanalysis.org>
- 33. W. Enck, P. Gilbert, B.-G. Chun, L. P. Cox, J. Jung, P. McDaniel, and A. N. Sheth, “TaintDroid: An Information-Flow Tracking System for Realtime Privacy Monitoring on Smartphones,” <http://static.usenix.org/event/osdi10/tech/slides/enck.pdf>
- 34. B. Gu, X. Li, G. Li, A. C. Champion, Z. Chen, F. Qin, and D. Xuan, “D2Taint: Differentiated and Dynamic Information Flow Tracking on Smartphones for Numerous Data Sources,” Technical Report, 2012.

Thank You